

Hints, Placebos and Shortcuts: Controlled Tests of Structural Primers for LLM Graph Reasoning

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Abstract

Prepending a *primer* of factual sentences about a graph’s structure — each node’s degree, clustering coefficient, random-walk return probability or component — to its text encoding is a natural way to help a language model answer questions about that graph. Across 84,000 generations from four Qwen3 arms (1.7B and 4B, thinking off and on) on six GraphQA tasks and seven edge densities over 40-node Erdős–Rényi graphs, we separate two things a primer can do: determine the answer by itself, which a graph-blind solver detects, and add text, whose cost we fit across five conditions spanning 37 to 4,691 characters. Length turns out to be close to free (a length term of -0.3 points per thousand characters, positive in 8 of 16 cells). Content that determines the answer is not: it is worth about $+16$ points where the model is failing and nothing where it is not, and on `node_degree` it costs qwen3-4b 12 points at $p=.50$ while gaining it 27 at $p=.85$ — same arm, same task, same primer, the sign following the accuracy the cell already had. The generated text shows why: the model stops counting and starts looking up, its median response falling from 273 tokens to 34. Structural side information that does not determine the answer is worth $+2$ to $+4$ points in the same window and nothing outside it, and more than half the cells of a standard sweep sit above 0.90 accuracy, where neither can show anything.

1 Introduction

Fatemi et al. (2023) show that a language model’s accuracy on graph-reasoning questions varies substantially with the text encoding of the graph. We ask whether prepending a *primer* — a short preamble of derived, node-level facts such as each node’s degree or clustering coefficient — improves the answers, and, when it changes them, what it changes.

Two things have to be separated before that question has an answer. A primer may help because the model reads the answer off it rather than reasoning

over the graph, which is the shortcut-learning concern of Geirhos et al. (2020) and the alternative to reasoning that Dziri et al. (2023) describe. And a primer may move accuracy simply by being there: added context is reported to degrade reasoning on its own (Shi et al., 2023; Levy et al., 2024). We address the first with a graph-blind solver that reads the primer text and never the graph, which says for each condition whether the primer determines the answer by itself; and the second by fitting the effect on primer length across five conditions spanning 37 to 4,691 characters, rather than trusting a single matched control.

Both separations change the conclusion. Length is close to free: fitted within each (arm, task), the length term averages -0.3 points per thousand characters and is positive in 8 of 16 cells. What is not free is content that determines the answer — and the surprise is that it frequently makes the answers worse. On `node_degree` the degree primer prints the requested number verbatim, and qwen3-4b falls from 99 to 87 points at $p=.50$, breaking 13 items for each one it fixes. The generated text shows the mechanism: the model stops counting and starts looking up, its median response falling from 273 tokens to 34.

The same primer, the same task and the same arm gain $+27$ points at $p=.85$, where that arm scores 33 without it. What decides the sign is neither the primer nor the task nor the model, but the accuracy the cell already had. Prompt-level graph interventions have been proposed before — Wang et al. (2023) give *Build-a-Graph* and *Algorithmic* prompting, Perozzi et al. (2024) learn a continuous encoding — and what we add is a controlled account of when one helps, when it hurts, and how much of the help is leakage, read from 84,000 generations on 40-node Erdős–Rényi graphs (Erdős and Rényi, 1959).

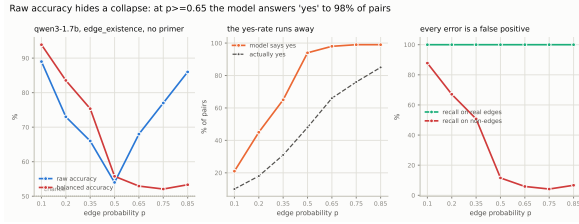


Figure 1: Why `edge_existence` needs balanced accuracy. `qwen3-1.7b` under `none`: raw accuracy falls to 54% at $p=.50$ and rises again to 86% at $p=.85$, which reads as a recovery. Balanced accuracy over the same responses reaches chance at $p=.65$ and stays there, because the model answers “yes” to 98–99% of pairs while 66–85% of them are edges.

2 Method

2.1 Primers and controls

A prompt is a primer, then the graph in the incident encoding, then the question. Seven conditions share one renderer, so they differ in content and never in format: `none`; four content primers reporting node degree, clustering coefficient, random-walk return probability (Dwivedi et al., 2022) and connected-component membership; `all`, which renders degree, clustering and `rwse` together; and `filler`. Mean primer lengths are 37, 891, 1,629, 2,949 and 4,691 characters for components, degree, clustering, `rwse` and `all`, and 1,829 for `filler`.

`filler` is a *degenerate-content* preamble of comparable length to `clustering`: forty sentences of the form “Node i is simply present within the graph G ”. It is not content-free — those sentences state the vertex set exactly, and a solver reading only `filler` scores 1.00 on `node_count` — and it is length-matched to `clustering` alone, so we use it as one condition among several rather than as the reference a content effect is defined against.

The control that does the work is the graph-blind solver. It parses the rendered primer text and applies exact rules, never touching the graph, so its score per (task, condition) is what the primer alone determines. We call a condition *answer-carrying* on a task when that bar is at least 0.85; the bars are bimodal — 18 of the 42 (task, condition) bars sit at exactly 1.000 and the highest of the rest is 0.746 — so any threshold between them gives the same split (Appendix Table 12).

2.2 Corpus, arms and measurement

Four arms answer one shared prompt set: `Qwen3-1.7B` and `Qwen3-4B` (Qwen Team, 2025), each with native thinking mode off (`qwen3-1.7b`, `qwen3-4b`) and on (`-think`). Graphs are Erdős–Rényi with $n=40$ at $p \in \{.10, .20, .35, .50\}$, 100 graphs per (task, density) cell, six GraphQA tasks and seven conditions, extended on `node_degree` and `edge_existence` to $p \in \{.65, .75, .85\}$. Decoding is greedy. The main sweep ran with an 8,192-token budget on every arm; the high-density extension ran at 2,048 on the plain arms and 8,192 on the thinking arms, so truncation rates are not comparable across those two bands and are reported separately (Appendix Table 13).

Three measurement decisions change what the numbers mean. *Exact match* is the outcome for every task; on `connected_nodes` the usual set-F1 sits at 0.93–1.00 in every cell while exact match spans 0.48–0.92, so F1 cannot separate the conditions. *Balanced accuracy* accompanies accuracy on `edge_existence`, where the share of node pairs that are edges rises from 10% to 85% across our density range: raw accuracy recovers with density while balanced accuracy falls to chance (Figure 2). And *truncation is an outcome*, not a missing value, so a pair is dropped when either side truncated — the convention least favourable to the primer — and rates are reported alongside. `node_count` and `cycle_check` have a constant gold answer at $n=40$ and hence a majority-class baseline of 1.000; they are reported in Appendix Table 3 onward but excluded from every summary over tasks. Differences are paired on the shared graph, tested with exact McNemar (McNemar, 1947) and corrected with Benjamini–Hochberg within each (arm, task) family. Extraction is not a confound: 99.84% of the 84,000 generations parse, and 99% of the 133 that do not are truncated, 128 of them on `cycle_check`.

3 Results

3.1 Most of a standard sweep cannot show an effect

A primer can only move a cell that is not already at floor or ceiling, and most cells are. Of the 324 side-information cells on the four informative tasks, 172 — 53% — sit above 0.90 accuracy without any primer and cannot show a gain by construction. Table 2 bins every cell by that baseline and splits it by whether the primer determines the answer. Both

Arm	Task	none	<i>n</i>	Δ vs none						Δ vs filler				
				filler	comp.	clust.	rwse	degree	all	comp.	clust.	rwse	degree	all
qwen3-1.7b	edge_existence	70.5	399	-15.7	-8.2	+3.8	+5.3	+4.0	+11.5	+7.5	+19.5	+21.1	+19.8	+27.2
	node_degree	60.5	399	-0.5	+1.8	+3.0	+4.0	+7.5	+10.3	+2.3	+3.5	+4.8	+8.0	+10.8
	connected_nodes	72.5	399	-6.2	+0.8	+1.3	+1.0	+3.5	+2.0	+7.0	+7.5	+7.3	+9.8	+8.3
	edge_count	1.8	211	-0.4	-1.4	-0.4	-0.8	+4.0	+2.4	-0.4	-0.4	-0.4	+4.5	+1.3
qwen3-4b	edge_existence	90.5	398	-7.0	-2.8	+4.0	-10.0	-4.3	-9.8	+4.2	+11.0	-3.0	+2.5	-2.8
	node_degree	99.2	400	-0.5	+0.2	-0.5	-2.3	-6.5	-8.0	+0.7	+0.0	-1.8	-6.0	-7.5
	connected_nodes	86.0	400	-12.5	+9.3	+1.0	+0.0	-3.0	-2.7	+21.8	+13.5	+12.5	+9.5	+9.8
	edge_count	2.0	393	+1.2	-1.3	-1.3	-1.5	+27.7	+3.8	-2.5	-2.5	-2.8	+26.5	+2.5
qwen3-1.7b-think	edge_existence	99.2	397	-0.8	-0.8	-1.0	-1.3	-2.3	-2.0	+0.0	-0.3	-0.5	-1.5	-1.2
	node_degree	76.8	376	-3.5	-1.0	+1.0	+2.5	+11.7	+11.6	+2.3	+4.5	+5.8	+14.6	+15.1
	connected_nodes	79.7	390	+2.0	+2.8	+1.5	-0.3	+7.4	+6.1	+0.5	-0.5	-2.5	+4.6	+4.1
	edge_count	86.3	<i>13</i>	+3.4	+7.4	-8.7	-4.0	-61.5	-8.3	+5.4	-4.2	+13.0	-18.8	-22.2
qwen3-4b-think	edge_existence	100.0	393	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	node_degree	99.7	380	+0.0	-0.3	-0.3	-2.1	+0.0	-0.8	-0.5	-0.3	-2.1	-0.3	-0.8
	connected_nodes	98.7	373	+0.8	+0.5	+0.0	+0.0	+0.0	-0.5	+0.0	+0.0	-0.5	-0.5	-0.5
	edge_count	97.6	<i>34</i>	-2.0	+1.7	+1.7	+0.0	-23.5	-41.5	+2.3	+0.0	-6.3	-19.3	-36.4

Table 1: Main sweep, $n=40$, pooled over densities $p \in \{0.10, 0.20, 0.35, 0.50\}$, 100 graphs per (task, primer, density) cell before truncation. The none column is exact-match accuracy (%); n is the smallest number of pairs in the row in which neither side truncated; the rest are paired differences in percentage points, against none on the left and against filler on the right, with truncated generations excluded pairwise. filler is length-matched to clustering only (Section 2.1), so the right half is a length-matched contrast for that column alone. node_count and cycle_check are omitted: gold is constant at $n=40$, so a shift there is not graph reading (Section 2.2); both appear in Appendix Table 3 onward. **Bold**: significant, exact McNemar, Benjamini–Hochberg within each (arm, task) family and baseline, $q = 0.05$. A row whose n is *italic* falls below 100 surviving pairs and is never bolded, because those survivors are selected on whether the generation terminated (Section 2.2).

columns show the same shape: nothing at the top of the range, and whatever movement exists concentrated between roughly 0.25 and 0.90; Appendix Table 11 counts the same effects by significance.

The two columns differ in size, not in shape. Inside that window a primer that carries the answer is worth about +16 points on average; genuine side information is worth +2 to +4. Above 0.90 both are indistinguishable from zero (-1.4 and -0.8). Per primer, inside the window, only all (+5.3), clustering (+2.9) and rwse (+2.2) are worth anything as side information; components averages -0.3, and its one large effect turns out not to be content at all (Section 3.3).

Thinking mode closes the window. The window is a property of the arm as much as of the task. Turning thinking on raises the median baseline from 56.5 to 79.0 on the 1.7B checkpoint and from 87.0 to 97.0 on the 4B one, which removes the cells a primer could have moved: of 110 cells per arm, 60 lie in the 0.25–0.90 window for qwen3-1.7b and only 15 for qwen3-4b-think. Within whatever window each arm has left, no primer is best twice — all is the best of the five on qwen3-1.7b (+9.8) and the worst on both 4B arms (-9.5, -13.0); degree leads on qwen3-1.7b-think (+12.3), clustering on qwen3-4b (+7.3) and components on

qwen3-4b-think (+27.7). A primer selected on one arm does not transfer to another (Appendix Figure 4).

Leakage is real but far from complete. Where the graph-blind solver scores 1.00 — the primer determines the answer outright — the model still has to find it, and mostly does not. On node_degree the recovered fraction runs from 0.99 on qwen3-4b-think down to 0.43 on qwen3-1.7b; on edge_count, where the answer requires summing the stated degrees rather than reading one of them, it runs 0.30 on qwen3-4b and 0.006 on qwen3-1.7b under all. A primer that contains the answer is not the same as a model that gives it, and the gap between the two is where the arms differ most.

3.2 The same primer helps and harms the same model on the same task

The cleanest demonstration needs no comparison across arms. On node_degree, qwen3-4b and the degree primer, which prints the queried node’s degree verbatim, the effect tracks the baseline across the seven densities: -6, -1, -7, -12 where the arm scores 99–100 without the primer, then -6 at 82, +7 at 50 and +27 at 33. Same arm, same task, same primer; the sign flips as the task gets hard enough for the model to need it.

Accuracy under	primer carries		side information	
none	cells	Δ	cells	Δ
0.00–0.25	19	+7.8	29	−1.8
0.25–0.50	9	+16.0	20	+2.3
0.50–0.75	9	+16.7	47	+3.9
0.75–0.90	4	−5.7	52	+0.6
0.90–1.00	28	−1.4	176	−0.8
<i>side information only, baseline 0.25–0.90</i>				
components	mean −0.3, median −0.5 over 28 cells			
clustering	mean +2.9, median +4.0 over 28 cells			
rwse	mean +2.2, median +3.0 over 28 cells			
degree	mean +2.0, median +1.5 over 18 cells			
all	mean +5.3, median +3.0 over 17 cells			

Table 2: Mean paired effect against `none` (percentage points) for every (arm, task, density, primer) cell of the main sweep, binned by the cell’s own `none` accuracy and split by whether the primer text alone determines the answer (shortcut bar ≥ 0.85 ; Table 12). Constant-gold tasks are excluded. Movement lives in the middle of the baseline range in both columns, and a primer that carries the answer is worth several times one that does not; above 0.90, where most cells sit, neither does anything.

The losses are not the cost of the added text. At $p=.50$ the primer breaks 13 items and fixes 1, and neither cell truncates at all. The response length says what happened: the median falls from 273 new tokens under `none` to 34 under `degree`, while `filler` — longer than `degree` — leaves it at 266 and costs 2 points. The model abandons a reliable count and reads a number off the primer instead, and the lookup is worse than the count.

3.3 What the primer changes in the response

The substitution is visible directly, and it pays when what it displaces was failing. On `edge_count` at $p=.50$, 7% of `qwen3-1.7b`’s responses under `none` state the handshake shortcut — sum the degrees, divide by two — and the rest enumerate edges. Under `degree`, which lists all 40 degrees, 100% state it, and truncation falls from 29% to 20%. At $p=.10$ the switch runs 67% to 99%. That gain is density-robust rather than a pooled average hiding one cell: +22, +30, +24, +33 across the four main densities.

The other robust gain we measure is not information transfer at all. `components` on `qwen3-4b/connected_nodes` runs +7, +11, +10, +9 from a 37-character preamble — but at $n=40$ every Erdős–Rényi graph at $p \geq .20$ is connected, so at three of those four densities the primer emits the same sentence about all

100 graphs, and the gain is larger there than at $p=.10$, where 52 of 100 graphs have more than one component and the string actually varies. Whatever that preamble buys, it is not graph-specific content. The same account covers the largest primer effects in the sweep: on `node_count`, whose gold answer is always 40, interposing almost any text between the encoder’s “nodes 0, ..., 39” preamble and the question interrupts an off-by-one that otherwise costs `qwen3-1.7b` 68.5% of its answers.

The direction of the length change is task-specific, so neither “primers make responses longer” nor its converse holds. On `qwen3-1.7b/edge_existence` the same primers make the model write more (206 tokens under `none` against 352 under `all`); on `qwen3-4b/connected_nodes` `filler` alone cuts the median from 233 to 94 (Appendix Table 9).

Beyond $p=.50$ on `edge_existence`, `qwen3-1.7b` stops reasoning altogether and answers “yes” to 98–99% of pairs. Neither prompt length nor task difficulty triggers this: a ladder that raises n at fixed mean degree makes the prompt longer and the task harder without producing it, and on the same graphs the model keeps working on the tasks whose answer is a number rather than a boolean. What the two-valued answer format permits is a cheap guess that is right most of the time — and raw accuracy rewards it while balanced accuracy does not (Figure 2; Appendix Figure 3).

3.4 Position and composition

Position inside the primer. The primer lists nodes in id order, so a retrieval account predicts that accuracy falls as the queried node moves down it. Several primers show steep gradients — `all` on `qwen3-1.7b-think` loses 24.3 points [−33.5, −15.2] from the first line to the last — and node identity is not the cause, since over 19,600 items a node’s id correlates +0.019 with its degree and +0.020 with its clustering coefficient. But `none` carries no primer and slopes too, by +25.0 points on `qwen3-4b/connected_nodes`, and `filler` has the same forty id-ordered lines as `degree`, so it is a second retrieval condition rather than a control. Testing each gradient against its own `none` control, only 6 of 48 are distinguishable. Position within the prompt moves accuracy; we cannot attribute it to the primer.

Bundling. `all` renders three primers together and is no shorter than their sum, yet across the 41 cells where the parts do anything it delivers a median 0.50 of what they are worth separately [0.29, 0.71]. A ratio of two noisy quantities is biased downward, so we simulate the matched null — every cell recentred on exact additivity, keeping its own noise scale and the same screen — which gives a median of 1.00 [0.92, 1.08] against the observed 0.50 ($p < 0.001$). On the 19 cells where no single part dominates — the only cells that separate dilution from selection — the ratio to the *sum* is 0.52 [0.40, 0.71] while the ratio to the *largest single part* is 1.00 [0.74, 1.20]. The bundle delivers what its best component delivers (Appendix Table 10).

4 Conclusion

Structural primers move GraphQA accuracy mainly by supplying the answer, and only where the model is not already producing it. Where a primer determines the answer it is worth about +16 points inside the difficulty window and nothing above it; genuine structural side information is worth +2 to +4 inside that window and nothing outside it. What the primer does, visible in the generated text, is offer a second procedure that the model takes: on `edge_count` the stated degrees replace enumeration with arithmetic in every response, and on `node_degree` the stated answer replaces a reliable count with an unreliable lookup, costing 12 points on a cell where the model was already right 99 times in 100.

Two measurements make this legible and neither is standard. Fitting the effect on primer length across conditions shows that length costs nothing here, so a single matched control cannot be what a content effect is defined against. And a graph-blind solver, scored per task and condition, separates the gains that are leakage from the gains that are not. Without both, the majority of a standard sweep — more than half of its cells sit above 0.90 — is uninformative about either.

Limitations

Two model sizes from one family, one encoding, one template, one primer position and greedy decoding throughout. All graphs are Erdős–Rényi, in which local clustering is nearly uniform, so the `clustering` primer carries little node-specific information. Most cells hold 100 graphs, which bounds what any single comparison establishes; the

claims above are counts and profiles over cells for that reason. The generation budget differs between the main sweep and the high-density extension, so truncation rates are not comparable across them.

Three limits bear on the behavioural results specifically. The procedure a response states is identified by a single surface pattern over free text (divided by 2, sum of the degrees, `handshak`); it is not hand-labelled, and a model may use a procedure it does not narrate, so those shares are the rate at which a response *says* it took the arithmetic route rather than the rate at which it did. Position within the primer is confounded with node identity, and the conditions available as controls carry the same positional structure, which is why we report that effect as unattributed. And the response-length collapse is attributed to the answer format by eliminating prompt length and difficulty, not by manipulating the format directly. Four conditions would settle the open questions: a shuffled-value primer as a stricter placebo, a corrupted primer to separate copying from computation, single-fact bundles to separate dilution from selection, and a permuted-order primer to break the position/identity confound.

Ethics Statement

This work evaluates open-weight language models on synthetically generated Erdős–Rényi graphs. It involves no human subjects, no personal or identifying data and no annotation labour. We see no direct risk of harm from the findings; the principal cost is the GPU compute for approximately 217,000 generations.

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A Supplementary material

A.1 Example prompt

A clustering-condition node_degree prompt at $p=0.10$ (elisions marked ...). Other conditions replace the first paragraph; none omits it.

Node 0 has clustering coefficient 0.20. Node 1 has clustering coefficient 0.00. ... Node 39 has clustering coefficient 0.00.

G describes a graph among nodes 0, 1, 2, ..., 38, and 39.

In this graph:

Node 0 is connected to nodes 2, 6, 8, 12, 35.

Node 1 is connected to nodes 5, 20.

...

Node 39 is connected to nodes 15, 16, 18, 20.

Q: What is the degree of node 16?

A:

Arm	none	filler	comp.	clust.	rwse	degree	all
1.7b	31.5	+39.8	+21.2	+67.3	+2.0	+6.8	+47.0
4b	90.2	+9.3	+9.0	+8.8	+9.3	+8.0	+9.8
1.7b-t	98.6	+1.4	+0.9	+1.4	+1.4	+0.8	+1.4
4b-t	99.5	+0.5	+0.0	+0.5	+0.5	+0.5	+0.5

Table 3: node_count, main sweep, pooled over $p \in \{0.10, 0.20, 0.35, 0.50\}$. The none column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. Gold is constant at $n=40$ on this task, so a shift need not reflect graph reading. **Bold:** exact McNemar, Benjamini–Hochberg within the arm, $q = 0.05$.

Arm	none	filler	comp.	clust.	rwse	degree	all
1.7b	99.7	+0.3	−0.6	+0.3	−7.5	−12.6	−1.1
4b	99.7	−0.3	−1.3	+0.3	+0.3	−1.5	−0.5
1.7b-t	100.0	+0.0	+0.0	+0.0	+0.0	−0.7	+0.0
4b-t	100.0	+0.0	+0.0	+0.0	+0.0	−0.8	+0.0

Table 4: cycle_check, main sweep, pooled over $p \in \{0.10, 0.20, 0.35, 0.50\}$. The none column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. Gold is constant at $n=40$ on this task, so a shift need not reflect graph reading. **Bold:** exact McNemar, Benjamini–Hochberg within the arm, $q = 0.05$.

Arm	none	filler	comp.	clust.	rwse	degree	all
1.7b	70.5	−15.7	−8.2	+3.8	+5.3	+4.0	+11.5
4b	90.5	−7.0	−2.8	+4.0	−10.0	−4.3	−9.8
1.7b-t	99.2	−0.8	−0.8	−1.0	−1.3	−2.3	−2.0
4b-t	100.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0

Table 5: edge_existence, main sweep, pooled over $p \in \{0.10, 0.20, 0.35, 0.50\}$. The none column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. **Bold:** exact McNemar, Benjamini–Hochberg within the arm, $q = 0.05$.

Arm	none	filler	comp.	clust.	rwse	degree	all
1.7b	60.5	−0.5	+1.8	+3.0	+4.0	+7.5	+10.3
4b	99.2	−0.5	+0.2	−0.5	−2.3	−6.5	−8.0
1.7b-t	76.8	−3.5	−1.0	+1.0	+2.5	+11.7	+11.6
4b-t	99.7	+0.0	−0.3	−0.3	−2.1	+0.0	−0.8

Table 6: node_degree, main sweep, pooled over $p \in \{0.10, 0.20, 0.35, 0.50\}$. The none column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. **Bold:** exact McNemar, Benjamini–Hochberg within the arm, $q = 0.05$.

Arm	none	filler	comp.	clust.	rwse	degree	all
1.7b	72.5	-6.2	+0.8	+1.3	+1.0	+3.5	+2.0
4b	86.0	-12.5	+9.3	+1.0	+0.0	-3.0	-2.7
1.7b-t	79.7	+2.0	+2.8	+1.5	-0.3	+7.4	+6.1
4b-t	98.7	+0.8	+0.5	+0.0	+0.0	+0.0	-0.5

Table 7: `connected_nodes`, main sweep, pooled over $p \in \{0.10, 0.20, 0.35, 0.50\}$. The `none` column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. **Bold**: exact McNemar, Benjamini–Hochberg within the arm, $q = 0.05$.

Arm	none	filler	comp.	clust.	rwse	degree	all
1.7b	1.8	-0.4	-1.4	-0.4	-0.8	+4.0	+2.4
4b	2.0	+1.2	-1.3	-1.3	-1.5	+27.7	+3.8
1.7b-t	86.3	+3.4	+7.4	-8.7	-4.0	-61.5	-8.3
4b-t	97.6	-2.0	+1.7	+1.7	+0.0	-23.5	-41.5

Table 8: `edge_count`, main sweep, pooled over $p \in \{0.10, 0.20, 0.35, 0.50\}$. The `none` column is exact-match accuracy (%); the rest are paired differences against it in percentage points, truncated generations excluded pairwise. **Bold**: exact McNemar, Benjamini–Hochberg within the arm, $q = 0.05$.

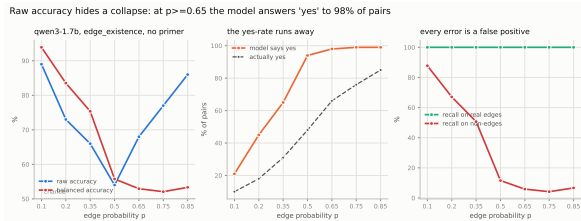


Figure 2: `edge_existence` under `none`, qwen3-1.7b. Accuracy falls to 54% at $p=0.5$ and then rises again to 86% at $p=0.85$, which reads as a recovery. Balanced accuracy over the same responses does not recover: it reaches chance at $p=0.65$ and stays there, because the model answers “yes” to 98–99% of pairs while 66–85% of them really are edges.

Arm / task		none	filler	degree	all
1.7b	median tokens	–	–	818	–
edge_count	% sum-of-deg.	7%	31%	100%	40%
	% truncated	29%	41%	20%	44%
4b	median tokens	273	266	34	134
node_degree	% sum-of-deg.	0%	0%	0%	0%
	% truncated	0%	0%	0%	0%
1.7b	median tokens	206	171	294	352
edge_existence	% sum-of-deg.	0%	0%	1%	0%
	% truncated	0%	0%	0%	0%
4b	median tokens	233	94	258	107
connected_nodes	% sum-of-deg.	0%	0%	0%	0%
	% truncated	0%	0%	0%	0%

Table 9: Response behaviour at $p=0.50$. Median new tokens is computed over generations that terminated, since a capped response is truncated at the budget rather than long, and is reported as – where more than 20% of generations truncated, because the surviving median is then a sample of the short responses; read the truncation row for those cells. “sum-of-deg.” is the share of responses that state the handshake shortcut (sum the degrees, divide by two) rather than enumerating edges. The primer moves length in both directions depending on the task, and on `edge_count` it replaces the method outright.

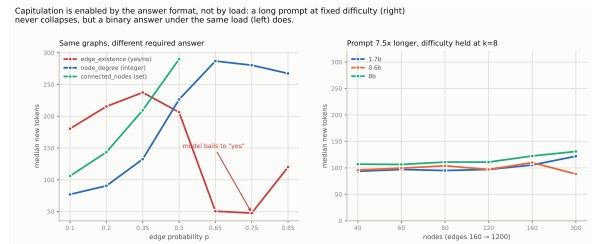


Figure 3: The response-length collapse on `edge_existence` and two controls on it. (a) median new tokens by density under `none` for three tasks answered about the *same* graphs: only the task with a two-valued answer collapses, while the tasks that must enumerate keep generating, so the trigger is not the graph. (b) median new tokens against n at fixed mean degree, where the prompt grows and the task gets harder without producing a collapse. What the collapse tracks is the answer format.

Arm	Task	$\sum$ parts	all	ratio
<i>other tasks</i>				
1.7b	connected_nodes	+5.7	+2.0	0.35
1.7b	edge_existence	+13.0	+11.5	0.88
1.7b	node_degree	+14.5	+10.2	0.71
1.7b-t	connected_nodes	+8.6	+6.1	0.71
1.7b-t	edge_existence	-4.5	-2.0	0.44
1.7b-t	node_degree	+15.6	+11.5	0.74
4b	edge_existence	-10.3	-9.8	0.95
4b	node_degree	-9.2	-8.0	0.86
<i>edge_count</i>				
1.7b-t	edge_count	-84.5	-22.2	0.26
4b	edge_count	+24.9	+3.8	0.15
4b-t	edge_count	-22.5	-41.1	1.82
<i>median over cells</i>				0.50

Table 10: all renders degree, clustering and rwse in one primer, so the parts and the bundle are both measured. The first two columns are paired differences against *none* in percentage points, averaged over $p \in \{0.10, 0.20, 0.35, 0.50\}$; rows whose parts sum to under 4 points are omitted, since a ratio of two null effects is not informative. The final row is the median over the 41 individual (arm, task, density) cells rather than over the averaged rows above, because averaging numerator and denominator over densities first gives a ratio of means that the largest cells dominate. Its bootstrap interval is $[0.29, 0.71]$. We read this as a direction and not as a coefficient: the cells are screened on a noisy denominator, and selecting on a noisy quantity inflates it, so a median of ratios sits below 1 even under exact additivity. The bias runs against the claim, so the bundle falling short of its parts survives it; its size does not (Section 3.4). *edge_count* is the widest case: its ratios run 0.26, 0.15, 1.82, from keeping almost none of the effect to overshooting it.

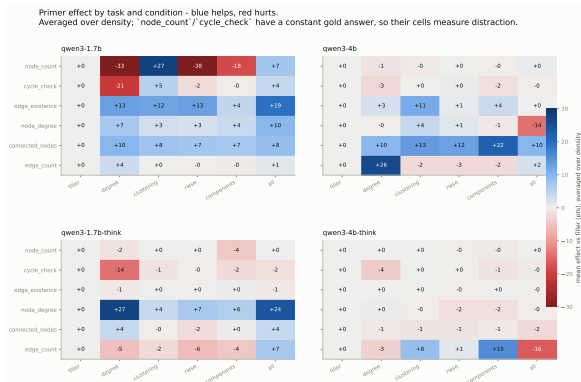


Figure 4: Mean paired effect against *filler* by task and primer, one panel per arm, on a shared symmetric scale so that zero is the same colour throughout. This covers all six tasks and all seven densities and scores truncated generations incorrect, so it is broader than Table 1 and its cells do not match it. The point of the figure is that the sign is not a property of a primer: rows and columns both change colour.

Accuracy without the primer (%)	significant effects cells	help	harm	median Δ (pts)
< 20	12	12	0	+23.0
20 – 50	12	11	1	+24.5
50 – 80	19	14	5	+18.0
80 – 95	7	2	5	-16.0
> 95	2	0	2	-13.5

Table 11: Every significant primer effect on the four non-constant-gold tasks, across all four arms and all densities, grouped by the accuracy that (arm, task, density) cell reached *without* a primer. Effects are paired differences against *none* in percentage points, exact McNemar with Benjamini–Hochberg correction, cells of at least 100 pairs. Below 80% a primer almost always helps and the median gain is around 20 points; above it a primer usually hurts, by a similar margin. Scoring truncated generations as wrong rather than dropping them changes the counts but not the crossover.

A.2 Per-task results

A.3 Response behaviour

A.4 Where effects live

Task	none	comp.	clust.	rwse	degree	filler	all
node_count	1.00	1.00	1.00	1.00	1.00	1.00	1.00
edge_count	.030	.030	.104	.030	1.00	.030	1.00
node_degree	.144	.144	.144	.210	1.00	.144	1.00
connected_nodes	.000	.000	.000	.001	.001	.000	.008
edge_existence	.735	.735	.727	.554	.746	.735	.741
cycle_check	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Table 12: Shortcut bars at $n=40$ (mean over seven densities): accuracy of a solver that reads only the rendered primer, never the graph. **Bold**: the primer content alone determines the answer. The unbolded 1.00 entries reflect a constant gold answer, not primer content.

A.5 Robustness analyses

Task	1.7B	1.7B-th.	4B	4B-th.
edge_count	22.4%	81.0%	0.6%	62.2%
cycle_check	6.1%	6.5%	1.0%	1.9%
node_degree	0.0%	1.1%	0.0%	1.6%
connected_nodes	0.0%	0.8%	0.0%	3.5%
edge_existence	0.0%	0.2%	0.1%	0.8%
node_count	0.0%	2.5%	0.0%	0.1%

Table 13: Truncation rate (share of main-sweep generations reaching the 8,192-token output budget), by task and arm, pooled over condition and density. **Bold**: above 50%.

A.6 Per-cell results

Paired effect sizes for every (arm, task, primer) cell: Δ in percentage points against *none*, an ex-

act 95% interval conditional on the discordant pairs (Clopper–Pearson, inverting the same exact McNemar test used for the per-cell p -values), the permutation p -value, and Δ under the alternative convention that scores truncated generations incorrect rather than dropping the pair (Section 2.2). n is the number of pairs in which neither arm truncated. “n/a” marks a cell in which no pair disagreed: the conditional estimate is then exactly zero with no width, which is a statement about this sample and not a bound on the effect, so the discordant count is reported instead.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	392	+2.8	[−0.7, +5.9]	0.1268	+2.5
	clustering	394	+1.5	[−2.7, +5.6]	0.5384	+2.0
	rwse	393	−0.3	[−4.3, +3.8]	1.0000	+0.2
	degree	390	+7.4	[+3.5, +10.6]	0.0001	+6.2
	filler	394	+2.0	[−1.7, +5.5]	0.3275	+2.5
	all	392	+6.1	[+2.2, +9.4]	0.0027	+6.2
cycle_check	components	346	+0.0	n/a (0 disc.)	1.0000	−7.5
	clustering	376	+0.0	n/a (0 disc.)	1.0000	−0.5
	rwse	386	+0.0	n/a (0 disc.)	1.0000	+2.2
	degree	295	−0.7	[−0.7, +0.5]	0.5036	−21.8
	filler	388	+0.0	n/a (0 disc.)	1.0000	+3.0
	all	373	+0.0	n/a (0 disc.)	1.0000	−0.8
edge_count	components	27	+7.4	[−5.1, +7.4]	0.5037	−0.8
	clustering	23	−8.7	[−17.2, +10.6]	0.6251	−1.2
	rwse	25	−4.0	[−4.0, +3.8]	1.0000	−3.2
	degree	13	−61.5	[−61.5, −16.1]	0.0102	−3.2
	filler	29	+3.4	[−8.4, +10.2]	1.0000	−1.5
	all	24	−8.3	[−22.8, +13.9]	0.6794	+5.0
edge_existence	components	398	−0.8	[−1.9, +0.9]	0.5090	−0.8
	clustering	398	−1.0	[−1.9, +0.6]	0.2911	−1.0
	rwse	398	−1.3	[−2.4, +0.6]	0.2274	−1.2
	degree	397	−2.3	[−3.1, −0.3]	0.0249	−2.5
	filler	399	−0.8	[−1.9, +0.9]	0.5051	−0.5
	all	399	−2.0	[−2.9, −0.1]	0.0374	−1.8
node_count	components	342	+0.9	[−0.6, +1.4]	0.3758	+1.5
	clustering	369	+1.4	[−0.1, +1.4]	0.0597	+9.0
	rwse	369	+1.4	[−0.1, +1.4]	0.0597	+9.0
	degree	360	+0.8	[−0.8, +1.8]	0.4512	+5.5
	filler	369	+1.4	[−0.1, +1.4]	0.0597	+9.0
	all	369	+1.4	[−0.1, +1.4]	0.0597	+9.0
node_degree	components	395	−1.0	[−4.4, +2.6]	0.6543	−1.0
	clustering	396	+1.0	[−3.0, +4.9]	0.7010	+1.5
	rwse	394	+2.5	[−1.9, +6.8]	0.2898	+2.5
	degree	376	+11.7	[+7.4, +15.0]	0.0001	+8.8
	filler	396	−3.5	[−7.3, +0.6]	0.0954	−3.0
	all	396	+11.6	[+7.3, +15.0]	0.0001	+12.2

Table 15: qwen3-1.7b-think: all cells.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	400	+0.8	[−3.1, +4.5]	0.7861	+0.8
	clustering	400	+1.2	[−3.3, +5.7]	0.6458	+1.2
	rwse	400	+1.0	[−3.5, +5.5]	0.7304	+1.0
	degree	400	+3.5	[−0.8, +7.5]	0.1239	+3.5
	filler	400	−6.2	[−10.0, −1.9]	0.0040	−6.2
	all	399	+2.0	[−2.9, +6.8]	0.4644	+2.0
cycle_check	components	343	−0.6	[−0.6, +0.4]	0.5033	+2.0
	clustering	362	+0.3	[−0.3, +0.3]	1.0000	+8.5
	rwse	358	−7.5	[−8.1, −5.2]	0.0001	−0.2
	degree	301	−12.6	[−12.6, −10.3]	0.0001	−17.5
	filler	341	+0.3	[−0.3, +0.3]	1.0000	+3.0
	all	362	−1.1	[−1.6, +0.5]	0.2210	+7.0
edge_count	components	211	−1.4	[−2.3, +1.0]	0.3783	−1.0
	clustering	238	−0.4	[−2.4, +1.9]	1.0000	−0.5
	rwse	239	−0.8	[−2.3, +1.4]	0.6837	−0.8
	degree	253	+4.0	[+0.6, +5.8]	0.0179	+3.0
	filler	224	−0.4	[−2.0, +1.6]	1.0000	−0.5
	all	211	+2.4	[−1.1, +4.6]	0.2269	+0.8
edge_existence	components	400	−8.2	[−10.9, −4.7]	0.0001	−8.2
	clustering	400	+3.8	[−0.9, +8.1]	0.1199	+3.8
	rwse	399	+5.3	[+0.4, +9.8]	0.0321	+5.0
	degree	400	+4.0	[−0.7, +8.4]	0.0957	+4.0
	filler	400	−15.8	[−18.3, −12.0]	0.0001	−15.8
	all	400	+11.5	[+6.5, +15.8]	0.0001	+11.5
node_count	components	400	+21.2	[+15.9, +25.6]	0.0001	+21.2
	clustering	400	+67.2	[+65.4, +67.2]	0.0001	+67.2
	rwse	400	+2.0	[−4.5, +8.5]	0.5871	+2.0
	degree	400	+6.8	[+2.9, +9.9]	0.0008	+6.8
	filler	400	+39.8	[+37.2, +40.6]	0.0001	+39.8
	all	400	+47.0	[+43.8, +48.7]	0.0001	+47.0
node_degree	components	400	+1.8	[−2.4, +5.8]	0.4599	+1.8
	clustering	400	+3.0	[−2.0, +7.8]	0.2577	+3.0
	rwse	399	+4.0	[−0.9, +8.6]	0.1104	+4.0
	degree	400	+7.5	[+2.8, +11.6]	0.0018	+7.5
	filler	400	−0.5	[−4.6, +3.7]	0.9028	−0.5
	all	400	+10.2	[+4.9, +15.0]	0.0003	+10.2

Table 14: qwen3-1.7b: all cells.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	400	+9.2	[+6.1, +11.3]	0.0001	+9.2
	clustering	400	+1.0	[−3.1, +5.0]	0.6893	+1.0
	rwse	400	+0.0	[−4.2, +4.2]	1.0000	+0.0
	degree	400	−3.0	[−7.1, +1.4]	0.1882	−3.0
	filler	400	−12.5	[−16.2, −7.9]	0.0001	−12.5
	all	400	−2.8	[−7.0, +1.7]	0.2511	−2.8
cycle_check	components	391	−1.3	[−1.8, +0.3]	0.1318	−2.0
	clustering	395	+0.3	[−0.2, +0.3]	1.0000	+0.5
	rwse	392	+0.3	[−0.2, +0.3]	1.0000	−0.2
	degree	391	−1.5	[−1.5, −0.1]	0.0312	−2.2
	filler	392	−0.3	[−0.3, +0.2]	1.0000	−0.5
	all	395	−0.5	[−1.0, +0.6]	0.6269	−0.2
edge_count	components	397	−1.3	[−2.1, +0.5]	0.1756	−1.2
	clustering	399	−1.3	[−2.4, +0.6]	0.2298	−1.2
	rwse	398	−1.5	[−2.0, +0.1]	0.0668	−1.5
	degree	393	+27.7	[+24.5, +29.4]	0.0001	+27.3
	filler	400	+1.2	[−0.9, +2.9]	0.3012	+1.2
	all	397	+3.8	[+1.1, +5.6]	0.0049	+3.8
edge_existence	components	400	−2.8	[−5.8, +0.7]	0.1254	−2.8
	clustering	400	+4.0	[+0.7, +6.7]	0.0168	+4.0
	rwse	400	−10.0	[−12.9, −6.2]	0.0001	−10.0
	degree	398	−4.3	[−7.3, −0.7]	0.0200	−4.8
	filler	400	−7.0	[−10.0, −3.3]	0.0002	−7.0
	all	399	−9.8	[−12.9, −5.8]	0.0001	−10.0
node_count	components	400	+9.0	[+6.9, +9.5]	0.0001	+9.0
	clustering	400	+8.8	[+6.2, +9.9]	0.0001	+8.8
	rwse	400	+9.2	[+6.9, +10.1]	0.0001	+9.2
	degree	400	+8.0	[+5.1, +9.7]	0.0001	+8.0
	filler	400	+9.2	[+7.5, +9.2]	0.0001	+9.2
	all	400	+9.8	[+8.0, +9.8]	0.0001	+9.8
node_degree	components	400	+0.2	[−0.2, +0.2]	1.0000	+0.2
	clustering	400	−0.5	[−1.7, +1.0]	0.7246	−0.5
	rwse	400	−2.2	[−3.1, −0.3]	0.0240	−2.2
	degree	400	−6.5	[−7.7, −4.0]	0.0001	−6.5
	filler	400	−0.5	[−1.7, +1.0]	0.7232	−0.5
	all	400	−8.0	[−8.9, −5.6]	0.0001	−8.0

Table 16: qwen3-4b: all cells.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	377	+0.5	$[-0.6, +1.0]$	0.6249	+0.5
	clustering	373	+0.0	$[-1.2, +1.2]$	1.0000	-0.8
	rwse	379	+0.0	$[-1.4, +1.4]$	1.0000	+0.5
	degree	376	+0.0	$[-1.5, +1.5]$	1.0000	+0.5
	filler	379	+0.8	$[-0.6, +1.3]$	0.3787	+1.2
	all	375	-0.5	$[-1.8, +1.1]$	0.7306	-1.2
cycle_check	components	381	+0.0	n/a (0 disc.)	1.0000	-3.2
	clustering	394	+0.0	n/a (0 disc.)	1.0000	+0.0
	rwse	395	+0.0	n/a (0 disc.)	1.0000	+0.2
	degree	375	-0.8	$[-0.8, +0.3]$	0.2514	-5.5
	filler	396	+0.0	n/a (0 disc.)	1.0000	+0.5
	all	392	+0.0	n/a (0 disc.)	1.0000	-0.5
edge_count	components	60	+1.7	$[-1.6, +1.7]$	1.0000	+24.8
	clustering	59	+1.7	$[-4.1, +5.0]$	1.0000	+24.8
	rwse	49	+0.0	$[-4.0, +4.0]$	1.0000	+10.2
	degree	34	-23.5	$[-23.5, -6.1]$	0.0073	+8.5
	filler	50	-2.0	$[-2.0, +1.9]$	1.0000	+12.2
	all	41	-41.5	$[-41.5, -25.3]$	0.0002	+1.8
edge_existence	components	396	+0.0	n/a (0 disc.)	1.0000	-0.5
	clustering	396	+0.0	n/a (0 disc.)	1.0000	-0.8
	rwse	393	+0.0	n/a (0 disc.)	1.0000	-1.2
	degree	398	+0.0	n/a (0 disc.)	1.0000	+0.0
	filler	399	+0.0	n/a (0 disc.)	1.0000	+0.0
	all	394	+0.0	n/a (0 disc.)	1.0000	-1.0
node_count	components	400	+0.0	$[-0.9, +0.9]$	1.0000	+0.0
	clustering	399	+0.5	$[-0.3, +0.5]$	0.4936	+0.2
	rwse	399	+0.5	$[-0.3, +0.5]$	0.5005	+0.2
	degree	399	+0.5	$[-0.3, +0.5]$	0.5005	+0.2
	filler	400	+0.5	$[-0.3, +0.5]$	0.4870	+0.5
	all	400	+0.5	$[-0.3, +0.5]$	0.4870	+0.5
node_degree	components	380	-0.3	$[-0.3, +0.3]$	1.0000	+0.2
	clustering	386	-0.3	$[-0.8, +0.6]$	1.0000	+1.8
	rwse	381	-2.1	$[-2.1, -0.5]$	0.0072	-1.2
	degree	388	+0.0	$[-0.5, +0.5]$	1.0000	+2.5
	filler	386	+0.0	n/a (0 disc.)	1.0000	+2.0
	all	383	-0.8	$[-0.8, +0.3]$	0.2466	+0.5

Table 17: qwen3-4b-think: all cells.